

**School of Arts and Sciences**

**Department of Computer Science and Information Technology**

Summer 2024 – 2025

**CSCI351 Concepts of Programming Languages**

**Class Day** : \_\_\_\_\_ **Class Time:** \_\_\_\_\_ **Room:** \_\_\_\_\_  
**Instructor** : \_\_\_\_\_  
**E-mail** : \_\_\_\_\_  
**Office** : \_\_\_\_\_ **Office Hours:** \_\_\_\_\_ **Extension No:** \_\_\_\_\_

**I. Course Description**

This course presents a study of human to computer communication. It is a survey of software development languages such that students would understand the concepts and design principles of programming paradigms (procedural, functional, and logical programming). Topics include a history of programming languages, data types supported, control structures, run-time management of dynamic programming structures, and much more.

**II. Course Objectives**

**CO-1:** Discover the fundamental concepts in programming language design and implementation.

**CO-2:** Show the primary constructs of programming language terminology and make correct use of them.

**CO-3:** Examine the properties of variables, methods, functions, and subprograms in programming languages.

**CO-4:** Develop simple programs in object-oriented programming languages such as C, C++, Java, Ruby, Python, etc.

**CO-5:** Evaluate the performance of a program written in different programming languages.

### III. Course Learning Outcomes

**CLO-1:** Discover the fundamental concepts in programming language design and implementation.

**CLO-2:** Show the primary constructs of programming language terminology and make correct use of them.

**CLO-3:** Examine the properties of variables, methods, functions, and subprograms in programming languages.

**CLO-4:** Develop simple programs in object-oriented programming languages such as C, C++, Java, Ruby, Python, etc.

**CLO-5:** Evaluate the performance of a program written in different programming languages.

### IV. Mapping to program outcomes

| Concept Of Programming Languages | PLO -1 | PLO -2 | PLO -3 | PLO -4 | PLO -5 | PLO -6 | PLO -7 | PLO -8 | PLO -9 |
|----------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| CLO-1                            |        |        |        | X      |        |        |        |        |        |
| CLO-2                            |        |        |        |        | X      |        |        |        |        |
| CLO-3                            |        |        |        | X      |        |        |        |        |        |
| CLO-4                            |        |        |        | X      | X      |        | X      |        |        |
| CLO-5                            |        |        |        |        |        | X      | X      |        |        |

### V. Topics Covered

- History of Programming Languages
- Control structures
- Dynamic Programming

### VI. Prerequisite – Corequisite

Pre courses

CSCI300 Intermediate Programming with Objects

### VII. Course Material:

Textbook: Concepts of Programming Languages 11<sup>th</sup> edition by Robert W. Sebesta

**Publisher:** Pearson. **ISBN-10:** 0-13-394302-X

### VIII. Course Assessment

| # Activity      | Percentage |
|-----------------|------------|
| Project I       | 10%        |
| Project II      | 10%        |
| Non common Test | 20%        |
| Midterm         | 25%        |
| Final Exam      | 35%        |

### IX. Detailed Schedule

| Week   | Book Reference                                              | Topic Details                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1 | Lecture 1<br>Concepts of Programming Languages 11th edition | Syllabus: reading and explaining.<br><b>Chapter 1: Preliminaries</b><br>1.1 Reasons for Studying Concepts of Programming Languages<br>1.2 Programming Domains Preliminaries<br>1.3 Language Evaluation Criteria<br>1.5 Language Categories<br>1.7 Implementation Methods<br>1.7.1 Compilation (brief)<br>1.8 Programming Environments                                                                         |
|        | Lecture 2<br>Concepts of Programming Languages 11th edition | <b>Chapter 3: Describing Syntax and Semantics</b><br>3.1 Introduction<br>3.2 The General Problem of Describing Syntax<br>3.3 Formal Methods of Describing Syntax<br>3.3.1 Backus-Naur Form and Context-Free Grammars (3.3.1.1 → 3.3.1.10)<br>3.4 Attribute Grammars<br>3.4.1 Static semantics<br>3.4.3 Attribute Grammars Defined<br>3.4.5 Examples of Attribute Grammars<br>3.4.6 Computing Attribute Values |
|        | Lecture 3<br>Concepts of Programming Languages 11th edition | <b>Chapter 3: Describing Syntax and Semantics</b><br>Solve Chapter Problem Set 3, 4, 5, 6, and 7 Pages 181-182<br>3.5 Describing the Meanings of Programs: Dynamic Semantics<br>3.5.1 Operational Semantics<br>3.5.1.1 The Basic Process<br>3.5.2 Denotational Semantics<br>3.5.2.1 Two Simple Examples                                                                                                       |
|        | Lecture 4<br>Concepts of Programming Languages 11th edition | <b>Chapter 3: Describing Syntax and Semantics</b><br>Solve Chapter Problem Set 11, 12, and 21 Pages 182-183<br>3.5.3 Axiomatic Semantics<br>3.5.3.1 Assertions<br>3.5.3.2 Weakest Preconditions<br>3.5.3.3 Assignment Statements<br>3.5.3.4 Sequences                                                                                                                                                         |

|        |           |                                                |                                                                                                                                                                                                                                                                                                                                                    |
|--------|-----------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 2 | Lecture 1 | Concepts of Programming Languages 11th edition | <b>Chapter 3: Describing Syntax and Semantics</b><br>3.5.3 Axiomatic Semantics<br>3.5.3.5 Selection<br>3.5.3.6 Logical Pretest Loops<br>3.5.3.7 Program Proofs                                                                                                                                                                                     |
|        | Lecture 2 | Concepts of Programming Languages 11th edition | Solve Chapter Problem Set 23, 24, 25, and 28 Pages 183-184<br><b>Chapter 5: Names, Bindings, and Scopes</b><br>5.1 Introduction<br>5.2 Names<br>5.3 Variables                                                                                                                                                                                      |
|        | Lecture 3 | Concepts of Programming Languages 11th edition | <b>Chapter 5: Names, Bindings, and Scopes</b><br>5.4 The Concept of Binding<br>5.4.1 Binding of Attributes to Variables<br>5.4.2 Type Bindings<br>5.4.2.1 Static Type Binding<br>5.4.2.2 Dynamic Type Binding<br>5.5 Scope<br>5.5.1 Static Scope<br>5.5.2 Blocks<br>5.5.3 Declaration Order<br>5.5.4 Global Scope<br>5.5.6 Dynamic Scope           |
|        | Lecture 4 | N/A                                            | <b>Chapter 5: Names, Bindings, and Scopes</b><br>5.6 Scope and Lifetime<br>5.7 Referencing Environments<br>5.8 Named Constants<br>Solve Chapter Problem Set 1, 6, 7, 8, 10, and 11 Pages 252-256<br><b>(Project I Distribution a group up to 3 students) Due on Week 4</b>                                                                         |
| Week 3 | Lecture 1 | Concepts of Programming Languages 11th edition | <b>Non-Common Test (Chapters 1 and 3)</b>                                                                                                                                                                                                                                                                                                          |
|        | Lecture 2 | Concepts of Programming Languages 11th edition | <b>Chapter 6: Data Types</b><br>6.1 Introduction<br>6.2 Primitive Data Types<br>6.2.1 Numeric Types<br>6.2.2 Boolean Types<br>6.2.3 Character Types<br>6.3 Character String Types<br>6.3.2 Strings and Their Operations<br>6.3.3 String Length Options<br>6.3.5 Implementation of Character String Types<br>6.4 Enumeration Types<br>6.4.2 Designs |

|        |           |                                                |                                                                                                                                                                                                                                                                                                                                                   |
|--------|-----------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | Lecture 3 |                                                | 6.5 Array Types<br>6.5.2 Arrays and Indices<br>6.5.3 Subscript Bindings and Array Categories<br>6.5.4 Array Initialization<br>6.5.5 Array Operations<br>6.6 Associative Arrays<br>6.6.1 Structure and Operations                                                                                                                                  |
|        |           | Concepts of Programming Languages 11th edition | <b>Chapter 6: Data Types</b><br>6.7 Record Types<br>6.7.1 Definitions of Records<br>6.7.2 References to Record Fields<br>6.7.4 Implementation of Record Types<br>6.8 Tuple Types<br>6.9 List Types<br>6.10 Union Types<br>6.11 Pointer and Reference Types<br>6.11.2 Pointer Operations<br>6.11.4 Pointers in C and C++<br>6.11.5 Reference Types |
|        |           | Concepts of Programming Languages 11th edition | <b>Chapter 7: Expressions and Assignment Statements</b><br>7.1 Introduction<br>7.2 Arithmetic Expressions<br>7.2.1 Operator Evaluation Order<br>7.2.2 Operand Evaluation Order<br>7.3 Overloaded Operators<br>7.4 Type Conversions<br>7.5 Relational and Boolean Expressions<br>7.6 Short-Circuit Evaluation<br>7.7 Assignment Statements         |
| Week 4 | Lecture 1 | Concepts of Programming Languages 11th edition | <b>Midterm (Chapters 1, 3, 5, and 6)</b>                                                                                                                                                                                                                                                                                                          |
|        | Lecture 2 | Concepts of Programming Languages 11th edition | <b>Project I Submission (each group should present its project)</b><br><b>Project II Distribution (a group up to 3 students)</b><br><b>Due on Week 5</b>                                                                                                                                                                                          |
|        | Lecture 3 | Concepts of Programming Languages 11th edition | <b>Chapter 7: Expressions and Assignment Statements</b><br>Solve chapter problem set 9, 10, 11, 13, and 19 Pages 350-352<br><b>Chapter 8: Statement-Level Control Structures</b><br>8.1 Introduction<br>8.2 Selection Statements<br>8.2.1 Two-Way Selection Statements<br>8.2.2 Multiple-Selection Statements                                     |

|               |                  |                                                |                                                                                                                                                                                                                                                                            |
|---------------|------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <b>Lecture 4</b> | Concepts of Programming Languages 11th edition | <b>Chapter 8: Statement-Level Control Structures</b><br>8.3 Iterative Statements<br>8.3.1 Counter-Controlled Loops<br>8.3.2 Logically Controlled Loops<br>8.3.3 User-Located Loop Mechanisms<br><br>Solve Chapter Programming Exercises 1, 3, 4, 7, and 8<br>Pages 386-387 |
| <b>Week 5</b> | <b>Lecture 1</b> | Concepts of Programming Languages 11th edition | <b>Project II Submission (each group should present its project)</b>                                                                                                                                                                                                       |
|               | <b>Lecture 2</b> | Concepts of Programming Languages 11th edition | Chapter 3 Problem Set Solve Exercises 6, 7, 8, 10, 11 and 12 Pages 181 & 182<br>Chapter 5 Problem Set Solve Exercises 6, 7, 9, 10, and 12 Pages 252, 253, 254, 255, 256, & 256                                                                                             |
|               | <b>Lecture 3</b> | Concepts of Programming Languages 11th edition | Chapter 6 Problem Set Solve Exercises 1, 7, 8 and 16 Pages 320 & 321<br>Chapter 7 Problem Set Solve Exercises 9, 11, 12 and 13 Page 349                                                                                                                                    |
|               | <b>Lecture 4</b> | Concepts of Programming Languages 11th edition | <b>Review for the final exam.</b>                                                                                                                                                                                                                                          |

#### **X. Class Rules & Discipline**

##### Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

##### Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the LIU student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

##### Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

### Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

### Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

### Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam's grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

### Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

### Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

### Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

### LIU E-mails

LIU students should use their LIU e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your LIU e-mail accounts.

### University Peer Review Policy

This is a new policy at LIU in which during the semester, all classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

### Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

## Equivalent Lettering Grades

| Percent (%) | QPTS | Letter Grade (G) | Percent (%) | QPTS | Letter Grade (G) |
|-------------|------|------------------|-------------|------|------------------|
| <60         | 0    | F                | 67          | 1.7  | C                |
| 60          | 1    | D                | 68          | 1.8  |                  |
| 61          | 1.1  |                  | 69          | 1.9  |                  |
| 62          | 1.2  |                  | 70          | 2    |                  |
| 63          | 1.3  |                  | 71          | 2.1  |                  |
| 64          | 1.4  |                  | 72          | 2.2  |                  |
| 65          | 1.5  | D+               | 73          | 2.3  | B+               |
| 66          | 1.6  |                  | 74          | 2.4  |                  |
| 75          | 2.5  | C+               | 83          | 3.3  |                  |
| 76          | 2.6  |                  | 84          | 3.4  |                  |
| 77          | 2.7  |                  | 85          | 3.5  |                  |
| 78          | 2.8  |                  | 86          | 3.6  |                  |
| 79          | 2.9  | B                | 87          | 3.7  |                  |
| 80          | 3    |                  | 88          | 3.8  | A                |
| 81          | 3.1  |                  | 89          | 3.9  |                  |
| 82          | 3.2  |                  | >=90        | 4    |                  |

| Grade | Description                               | QPTS |
|-------|-------------------------------------------|------|
| A     | Excellent                                 | 4.00 |
| B+    | Very Good                                 | 3.50 |
| B     | Good                                      | 3.00 |
| C+    | Above Average                             | 2.50 |
| C     | Average                                   | 2.00 |
| D+    | Fair                                      | 1.50 |
| D     | Poor                                      | 1.00 |
| F     | Fail                                      | 0.00 |
| IP    | Course in Progress                        |      |
| I     | Incomplete work                           |      |
| W     | Student has withdrawn from the course     |      |
| AW    | Administrative withdrawal from the course |      |
| R     | Course has been repeated                  |      |
| S     | Satisfactory work                         |      |
| U     | Unsatisfactory work                       |      |
| %G    | Final Grade                               |      |
| QPTS  | Quality Points                            |      |